

[Titel]

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August 20, 2026

Abstract

Experimental evidence suggests that delegation shifts responsibility for adverse outcomes onto intermediaries. We ask whether the same responsibility-avoidance motive applies to delegation to algorithms. In a preregistered online experiment with 322 participants, a decision-maker chooses to make a payoff-relevant prediction for a matched recipient or delegate it to an algorithm, calibrated to perform equally well. We vary between subjects whether the recipient can impose punishment. The prospect of punishment reduces delegation by 17 percentage points, from 59 to 43% reversing the direction that the responsibility-shifting account predicts. Delegation offers no protection in return. Recipients punish delegated and self-made predictions alike and condition punishment on the realized outcome. We discuss several candidate mechanisms for this reversal.

JEL Classification: C91, D91, O33

Keywords: algorithms; delegation; responsibility; punishment; experiment

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1 Introduction

[Unify H1, R1, Hypothesis 1, Result1 notation.]

Algorithms take part in a growing share of consequential decisions. Many of these decisions carry consequences that fall, at least in part, on others. Physicians rely on diagnostic systems, firms let screening software rank job applicants, and financial advisors draw on algorithmic tools. In each of these settings, a human decides whether to put the algorithm in charge. Who bears responsibility when an algorithm’s decision causes harm is the subject of a lively legal (Lemley and Casey, 2019; Buiten et al., 2023), ethical (Matthias, 2004; Santoni de Sio and Mecacci, 2021), and policy (Expert Group on Liability and New Technologies – New Technologies Formation, 2019; European Union, 2024) debate. While the debate concerns where responsibility should lie, behavior is another matter. Does delegating a decision to an algorithm change the responsibility that those affected assign to the delegator? And does the prospect of being held responsible move the delegation decision itself? This paper provides experimental evidence on both.

For delegation between humans, the experimental literature points in a clear direction. Delegation shifts responsibility. Principals who delegate an unpopular decision to a human intermediary are punished less for it, feel less responsible for it, and delegate strategically for that reason (Bartling and Fischbacher, 2012; Hamman et al., 2010; Coffman, 2011; Oexl and Grossman, 2013). Recent evidence suggests that algorithms can inherit the intermediary’s role. Delegating to an algorithm helps decision-makers escape blame for adverse outcomes (Feier et al., 2022), lets them shed moral responsibility for self-serving choices (Tontrup and Sprigman, 2025) and makes dishonesty easier by obscuring intent (Köbis et al., 2025). If algorithms provide such cover, decision-makers who face potential punishment should be drawn to them, and delegation should protect the delegator when outcomes turn out badly. Yet no existing design varies whether punishment is possible, so whether the prospect of being held responsible in fact motivates delegation remains untested.

We address both questions in a preregistered online experiment with 322 participants.¹ A decision-maker, Player A, first completes ten rounds of a prediction task for her own payoff, and then makes one further prediction that determines the payoff of a matched recipient, Player B. Player A has the option to either make the consequential prediction herself or delegate it to an algorithm that has been calibrated to perform equally well at the task. The environment we study, a prediction under genuine uncertainty made by a decision-maker with no material stake in the affected party’s outcome, resembles the settings in which algorithmic delegation typically operates in practice. We then vary between subjects whether Player B can reduce Player A’s earnings. In the *Punishment* condition, we elicit Player B’s punishment for every combination of delegation choice and prediction outcome, providing a within-subject test of whether delegation insulates Player A from punishment, conditional on the outcome. In the *No-*

¹The pre-registration (AsPredicted #133980) is available at <https://aspredicted.org/hs9az8.pdf> and reproduced in full in Appendix C.

Punishment condition, no such punishment is possible. With everything except the possibility of punishment held fixed, any difference in delegation rates is attributable to that possibility, allowing us to test directly whether the prospect of being held responsible moves the delegation decision.

We find that the prospect of punishment reverses the canonical prediction. When punishment is possible, 42.5 percent of decision-makers delegate to the algorithm. When it is not, 59.3 percent do. The possibility of punishment thus *reduces* delegation by roughly 17 percentage points. Delegation, in turn, offers no protection against punishment. Conditional on the outcome, Player B punishes delegated and self-made decisions alike, and punishment responds almost entirely to the outcome of the prediction rather than to how it was made.

The possibility of punishment thus moves the delegation decision even though delegating draws no additional punishment. We examine several explanations to reconcile these findings. The first is anticipated punishment. Player A may keep the decision because she expects delegation to be punished more harshly. Her elicited beliefs show no such expectation, and instead match how Player B actually punishes. The second is punishment-induced effort. Player A may keep the decision in the hope of outperforming the algorithm in the prediction task for Player B, in which case non-delegators should improve on their earlier performance precisely when punishment is possible. Instead, improvement concentrates where punishment is impossible. What the data do show is selection. When punishment is possible, it is disproportionately the better-performing Player As who keep the decision, and the delegation gap between conditions widens with performance. Both an intrinsic preference for producing the good outcome oneself and a distaste for delegating uncertain decisions are consistent with this selection, but they operate in both conditions alike and cannot on their own generate the treatment difference. Lastly, by keeping the decision, Player A may aim to convey to Player B that she intends to try especially hard on his behalf, whereas delegating could be perceived as shirking the decision. Because the task is inherently uncertain, sending this signal requires no measurable additional effort. However, the signal carries monetary value only if Player A expects apparent shirking to draw harsher punishment, which her beliefs do not show. Detached from actual punishment, she may instead care about Player B's judgment itself, although this too explains the treatment difference only if the prospect of punishment is what activates this motive. Thus, while the drop in delegation under punishment is robust, which motive produces it is a question we narrow rather than settle.

Our contribution is twofold. First, we provide the first direct test of whether the prospect of punishment motivates delegation to an algorithm, the motive the responsibility-avoidance literature identifies for human intermediaries (Bartling and Fischbacher, 2012; Hamman et al., 2010; Coffman, 2011; Oexl and Grossman, 2013). Our environment departs from these canonical designs in three respects. The intermediary is an algorithm rather than a person. The decision involves genuine uncertainty, so a bad outcome may reflect bad luck rather than bad intent. And its consequences fall entirely on the recipient, so there is no self-serving choice to police.

In this environment, the motive operates with the opposite sign. Accountability disciplines algorithm use rather than fueling it, and it does so selectively among the decision-makers most likely to succeed. This reversal is the paper’s main empirical result, adding a determinant of algorithm use that operates through the decision context rather than through the algorithm or the task.

Second, we document how an affected party assigns responsibility for algorithmic delegation when the delegation choice is uninformative about relative ability. The closest precedent is [Feier et al. \(2022\)](#), whose treatments vary the type of delegate under a fixed punishment regime, whereas we hold the delegate fixed and vary the punishment regime. In their environment, decisions delegated to an algorithm are rewarded more generously than self-made ones after bad outcomes. However, the same subject both delegates and evaluates, and the delegation decision is strongly determined by beliefs about relative ability, inviting punishment of overconfidence rather than of delegation itself. Once both channels are closed, the differential disappears, and punishment tracks the realized outcome instead.

The remainder of the paper is organized as follows. Section 2 relates the study to the literature. Section 3 presents the experimental design and hypotheses. Section 4 reports the results and examines candidate mechanisms. Section 5 concludes.

2 Related Literature

Responsibility avoidance through delegation

Delegation has long been studied as an efficiency-enhancing institutional arrangement in which principals rely on agents with better information, lower opportunity costs, or specialized skills.² Alongside these efficiency motives, the literature has identified a range of strategic motives. Delegation can serve, for instance, as a commitment device that alters the equilibrium ([Schelling, 1960](#); [Fershtman and Gneezy, 2001](#)), as a way to motivate the agent ([Aghion and Tirole, 1997](#)), or as a means of shifting responsibility for the outcome away from the principal, both in her own eyes (felt responsibility) and in the eyes of others (attributed blame).

The political-science and public-choice literatures recognized early that delegation can shift blame onto an intermediary ([Weaver, 1986](#); [Fiorina, 1986](#)). Experimental economics has established the same motive in incentivized settings, primarily using modified dictator games. [Bartling and Fischbacher \(2012\)](#) provide the canonical experimental evidence. In their design, a dictator may either choose between a fair and an unfair allocation of an endowment or instead delegate this choice to another person whose monetary incentives are aligned with the dictator’s. A recipient, who is adversely affected by the unfair allocation, may then assign costly punishment to the dictator and the intermediary. They find that delegation effectively shifts punishment from the dictator to the intermediary, and the prospect of avoiding punishment con-

²For an overview of the principal-agent literature see, e.g., [Bolton and Dewatripont \(2005\)](#).

stitutes a strong motive for the delegation decision itself. Complementary experiments suggest that the reduction in the principal’s punishment stems from the absence of direct interaction with the affected party rather than from a pure transfer of blame to a deciding intermediary. The principal escapes punishment even when the intermediary remains completely passive, but not when the principal continues to interact with the recipient directly (Coffman, 2011). The blame the intermediary absorbs, by contrast, attaches to the act of choosing rather than to any contribution to the outcome. Punishment lands on the intermediary even when delegation itself has eliminated the fair option, leaving him only the choice between two unfair allocations (Oexl and Grossman, 2013). Consistent with blame attaching to the act of choosing, randomization devices provide the principal weaker cover than deciding humans (Bartling and Fischbacher, 2012; Oexl and Grossman, 2013) and attract correspondingly less delegation (Bartling and Fischbacher, 2012). The responsibility-shifting motive does not require a punishment stage. Hamman et al. (2010) find that principals’ transfers to recipients fall sharply under delegation even without any punishment opportunity, and that principals report feeling less responsible for the resulting allocations, indicating that responsibility shifting also operates through felt responsibility and moral wiggle room.

Beyond the canonical dictator-game setting, the responsibility-shifting motive, and the strategic use of delegation it gives rise to, have been documented across a range of domains. Voters strategically avoid being pivotal, and non-pivotal voters are indeed punished less for the same group outcome (Bartling et al., 2015). Principals pay agents to lie on their behalf and deceive more readily through a delegate than directly (Erat, 2013; Kandul and Kirchkamp, 2018; Gawn and Innes, 2019). Observers rate delegating legislatures as less blameworthy, and do so even when the delegate is powerless to change the outcome (Hill, 2015). Freer et al. (2024) isolate the motive in financial decision-making, finding that a substantial fraction of investors delegate even purely luck-based lottery choices, in which alternative motives such as decision costs, performance-chasing, or risk tolerance cannot rationalize the choice.

In sum, the canonical literature documents that, in settings with self-interested choice, delegation reduces both the principal’s punishment and her felt responsibility, in turn motivating delegation. Our setting departs from this canonical design along three dimensions. First, the intermediary is algorithmic rather than human. Second, the outcome falls entirely on a passive recipient rather than partly on the principal herself. Third, the underlying decision involves genuine predictive uncertainty rather than a known fair–unfair choice, removing the transparent self-serving intent that punishment sanctions in the canonical designs. Whether the responsibility-shifting motive retains its force under these conditions is the question we take up.

Delegation to algorithms

A large empirical literature studies when and why people are willing to use algorithmic decision aids. Dietvorst et al. (2015) coin the term *algorithm aversion* for the finding that people prefer

their own or other humans' judgment over algorithmic predictions, even after observing that algorithms outperform human decision-makers.³ In contrast, Logg et al. (2019) document conditions under which subjects show *appreciation* of algorithmic advice over human advice. The picture that emerges is therefore one of substantial heterogeneity in algorithm preferences across tasks, contexts, and elicitation methods. Burton et al. (2020), Mahmud et al. (2022), Chugunova and Sele (2022), Qin et al. (2025) and Irlenbusch (2026) provide systematic reviews of this literature and document a large array of moderators for when and why algorithms are used in decision-making.

Some of these moderators attach to the algorithm, to the task, and to the authority the human retains. On the algorithm side, aversion is stronger when the algorithm cannot be observed to learn from its mistakes (Berger et al., 2021) or remains a black box to its users (Litterscheidt and Streich, 2020). Evidence about relative performance gathered through experience and feedback reduces aversion (Filiz et al., 2021), whereas explanations of how the algorithm works leave acceptance unchanged (Dargnies et al., 2026). On the task side, aversion rises with the degree of irreducible uncertainty (Dietvorst and Bharti, 2020) and with the perceived subjectivity of the task (Castelo et al., 2019), while reliance on algorithmic advice increases with task difficulty (Bogert et al., 2021) and under time pressure (Jung and Seiter, 2021). On the authority side, people use imperfect algorithms far more readily when they retain even a minimal ability to modify the output (Dietvorst et al., 2018), experimental firms likewise delegate pricing more often when they can override the algorithm (Normann et al., 2025), and part of measured aversion reflects a reluctance to relinquish decision authority as such rather than distrust of algorithms (Ivanova-Stenzel and Tolksdorf, 2025, 2024). Authority, in turn, may shape where responsibility lands. The mere existence of a manual override may increase the blame humans receive when the machine errs (Arnestad et al., 2024), and experienced professionals, who answer for outcomes in their domain, can become particularly averse to algorithmic advice (Allen and Choudhury, 2022). Much of this literature concerns advice-taking, in which the decision-maker retains final authority over the outcome. We instead study full delegation, in which the final prediction is ceded to the algorithm entirely.

Preferences for or against algorithms as such produce level effects on delegation. Our between-condition comparison holds the algorithm, the task, and the authority structure fixed and varies only whether punishment is possible, so that such level effects difference out.

Responsibility perceptions of decisions involving algorithms

Should we include the Arnestad and Allen references down here maybe?

Within this body of work, a smaller subset focuses on algorithm use preferences in settings whose consequences fall partly or wholly on someone other than the decision-maker, sometimes

³Interest in the comparison between human and statistical judgment is long-standing. See Meehl (1954) and Dawes (1979) for the classic demonstrations that simple statistical rules outperform clinical judgment and are resisted nonetheless.

called the *moral domain* (Gogoll and Uhl, 2018). Much of this work concludes that aversion to algorithms is amplified once third-party welfare is at stake. Gogoll and Uhl (2018) document this in an incentivized calculation task with performance uncertainty, Bigman and Gray (2018) in vignettes across high-stakes moral domains (driving, legal, medical, military), and Jauernig et al. (2022) identify moral discretion as the human capacity that people are most reluctant to hand over in such contexts.⁴

Once decisions affect a third party, the responsibility-shifting motive established for delegation to humans becomes relevant for algorithmic delegation as well, and with it the question of how observers apportion blame when an algorithm is involved in the decision. A first strand of evidence, predominantly vignette-based and unincentivized, finds leniency toward the human decision-maker. Physicians who follow standard-of-care AI recommendations are judged less harshly, and face lower legal exposure, than physicians who reject them, among lay and expert evaluators alike (Tobia et al., 2021; Tacconelli et al., 2026). Algorithmic discrimination in hiring triggers less moral outrage than identical human discrimination (Bigman et al., 2023). In autonomous driving, where such attributions have been studied most extensively, judgments cut both ways and track the allocation of control. Where the vehicle decides, responsibility moves to the machine, and people judge the same harm to a pedestrian more permissible when the autonomous vehicle causes it than when they cause it themselves (Gill, 2020). Where control is shared, blame stays with the human. Drivers attract more of it than their vehicles when both err (Awad et al., 2020), even for crashes they cannot reasonably avoid (Beckers et al., 2022). The division-of-authority pattern extends beyond driving, with humans who merely monitor an algorithm blamed less than humans who decide alone (Shank et al., 2019).

A second strand uses incentivized experiments. Much of it finds that algorithmic intermediaries can serve as moral cover. Chevrier and Teixeira (2024) extend the canonical dictator game of Bartling and Fischbacher (2012) with an algorithmic intermediary. As with a human intermediary, the dictator who delegates an inequalitarian allocation faces lower expected punishment than the dictator who chooses it herself. Delegation to the algorithm makes recipients less likely to punish at all, although the delegator is punished more harshly than under a human intermediary when punishment does occur. Delegation rates in their design nevertheless do not differ between human and algorithmic intermediaries. In a similar setting, Tontrup and Sprigman (2025) let dictators hand the execution of their transfer to a large language model. Transfers fall when this option is available, and dictators as well as outside observers attribute part of the responsibility to the machine. Köbis et al. (2025) find that delegation to algorithms increases dishonest behavior, and in an incentivized donation choice with real moral stakes, Hüholt and Szech (2026) find that subjects delegate to an AI more often than to a human counterpart and feel less responsible when they do. The cover has limits, however. Ismagilova and Ploner (2025) let agents invest on behalf of principals who can then reward or punish them. Rewards and punishments follow the outcome, and they do so no differently for delegated and

⁴For recent surveys of this area, see Bonnefon et al. (2024) and, with a legal-policy focus, Sunstein and Gaffe (2024).

self-made decisions.

The study closest to ours is [Feier et al. \(2022\)](#). In their design, a subject determines the payoff of another participant either through her own performance in a Raven-style logic task or, depending on the treatment, through delegation to another human or to an algorithm. The outcome of this choice falls entirely on the other participant, who can reward or punish the subject for the decision and its result. They find that after good outcomes delegated and self-made decisions are rewarded alike in both treatments. After bad outcomes, decisions delegated to the algorithm, but not decisions delegated to another human, are rewarded more generously than self-made ones. The delegation decision itself is similar across the two treatments.

Our design departs from theirs in three respects. First, their algorithm is calibrated at the session level, so the delegation decision signals the subject’s beliefs about her own ability relative to the session performance distribution. As a result, punishment may partly sanction perceived overconfidence, outweighing any downstream punishment as a measure of responsibility shifting.⁵ We calibrate the algorithm at the individual level, which removes the relative-performance signal from the delegation decision by construction. Second, in their design the same subject acts as both delegator and evaluator, which can import motives of self-justification or hypocrisy aversion into the punishment decision.⁶ We separate the two roles across disjoint subject pools. Third, their design holds the punishment possibility fixed across treatments, so it cannot identify whether the prospect of punishment motivates delegation in the first place. Any difference in delegation rates may be due to general preferences for algorithm use. We vary the punishment possibility directly.

Our experiment is thus, to our knowledge, the first direct test of responsibility shifting onto an algorithm in a setting in which the outcome is genuinely uncertain and delegation is uninformative about relative ability. It is also the first to identify the effect of the prospect of punishment itself on the decision to delegate to an algorithm.

3 Experimental Design

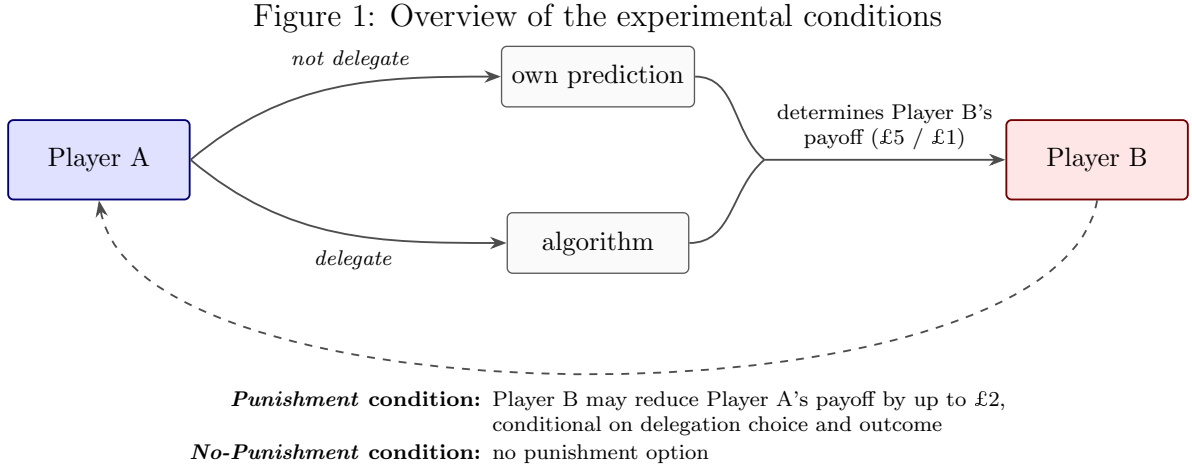
We ran an online experiment via the British platform Prolific (<https://www.prolific.com>) with UK participants. The experiment was programmed in oTree ([Chen et al., 2016](#)) and eligibility followed Prolific’s standard quality screens, a minimum 95% prior approval rate, a fluent-English filter, and a desktop-only restriction (no mobile or tablet) that kept the screen layout consistent across subjects.

The design combines a between-subject manipulation of whether costly punishment is possible with a within-subject elicitation of conditional punishment behavior. Within each condition,

⁵Their own analysis shows that the propensity to delegate is driven primarily by beliefs about task performance, not by the type of delegate.

⁶In their data, evaluators who themselves delegated evaluate delegated decisions more favorably than self-made ones after good outcomes, whereas non-delegating evaluators favor self-made decisions ($r = -0.34$, $p = 0.004$, machine treatment; own computations from their published data).

participants assumed one of two roles. Player A decides whether to make a payoff-relevant prediction for Player B herself, or to delegate the prediction to an algorithm calibrated to match her own measured performance. Player B is the matched recipient whose payoff is fully determined by the outcome of the prediction, whether Player A makes it herself or delegates it to the algorithm. In the *Punishment* condition Player B may subsequently punish Player A, conditional on both the delegation decision and the realized outcome, while in the *No-Punishment* condition no punishment is possible. Figure 1 summarizes the basic setup.



Procedure

Player A

After providing informed consent, Player A completed ten incentivized rounds of a visual prediction task. Specifically, in each round, she predicted a person's weight from a facial image and the person's height. She could enter predictions via a slider or type them directly in kilograms, pounds, or stones-and-pounds.⁷ No feedback was provided between rounds. At the end of the experiment, one of the ten rounds was randomly selected for payment. Player A earned £4 if her prediction in that round was within 10 lbs of the true weight and £2 otherwise. These initial rounds served three purposes simultaneously: First, they established a stake from which any later punishment could be deducted. Second, they familiarized Player A with the task. Third, they generated the performance data needed to calibrate the algorithm.

The choice of a visual prediction task is not incidental. Algorithm preferences may depend sensitively on the algorithm's perceived relative competence (Qin et al., 2025; Logg et al., 2019). Our design neutralizes this concern by calibrating the algorithm to match each subject's individual performance. The visual nature of the task lends credibility to this equal-performance claim, as people have everyday familiarity with judging others' build and appearance, and algorithmic error is plausible.⁸ In contrast, subjects may be skeptical that an algorithm could

⁷All individuals depicted in the images had explicitly consented to the use of their photographs for research purposes.

⁸Predicting body weight from an image has been used in earlier experiments (Logg et al., 2019).

ever fail in purely mathematical tasks (Gogoll and Uhl, 2018; Feier et al., 2022). Finally, we chose a prediction task involving genuine uncertainty rather than a setting with an obvious self-serving motive (as in classic dictator-game variations).⁹ This reflects the nature of many real-world applications of algorithms, in which outcomes are inherently uncertain, the decision-maker has no direct material interest in the affected party’s outcome, and performance is evaluated ex post.

After the ten rounds, we randomly paired Player A with another participant, Player B, who had not completed the prediction task himself. We informed Player A that her final prediction would not affect her own payoff but would almost entirely determine Player B’s. A prediction within 10 lbs of the true weight would give Player B a high payoff (£5), and any less accurate prediction a low payoff (£1).¹⁰ Player A then learned that she could either make the prediction herself or delegate it to an algorithm. She was also told that, based on her first ten predictions, the algorithm would have performed equally well.

We calibrated the algorithm to mimic Player A’s own performance over the first ten rounds. Concretely, if Player A delegated the final prediction to the algorithm, the algorithm was equivalent to drawing a Bernoulli outcome with success probability equal to Player A’s empirical accuracy rate over her first ten rounds. For example, a subject who scored 3 out of 10 in the initial rounds thus delegates to an algorithm that produces the high payoff for Player B with 30% probability.

Fixing the algorithm’s success probability to each subject’s own accuracy makes the equal-performance claim hold subject by subject, not merely on average across a pool of participants. This matters because beliefs about relative performance are otherwise likely a dominant driver of delegation.¹¹ For example, under the session-level calibration of Feier et al. (2022), the delegation choice is largely driven by each subject’s sense of her ability relative to the algorithm. Matching the algorithm to each subject neutralizes this motive, and making this performance parity common knowledge should limit Player B from reading delegation as a signal of Player A’s believed relative competence. This narrows a signal-extraction channel that may otherwise contribute to Player B’s punishment.

In the *Punishment* condition, we further informed Player A that Player B could impose (costly) punishment of up to £2, and could condition punishment on both her delegation decision and the realized outcome. In the *No-Punishment* condition, Player B had no such punishment option. The two delegation options (own decision / algorithm) appeared in random order across subjects. We attached no monetary cost to delegation. This brings us closer to Player A’s

⁹With no self-serving option available, an intention-based punishment motive should have less bite than in those settings.

¹⁰Since the final prediction determines only Player B’s payoff, Player A has no financial interest in Player B receiving the low payoff. As long as a better outcome for Player B weakly reduces Player A’s expected punishment in the *Punishment* condition, the incentives of the two players can be considered aligned.

¹¹This motive would not inherently bias the treatment-effect comparison, since it should be balanced across the *Punishment* and *No-Punishment* conditions, but it might otherwise dominate the delegation decision and may obscure the responsibility-shifting motive of interest.

intrinsic preference over delegation in the *No-Punishment* condition, since with any positive cost, the responsibility-shifting motive would need to outweigh a price effect. Thus, costless delegation is also standard in the literature (Bartling and Fischbacher, 2012; Coffman, 2011; Feier et al., 2022). If Player A chose to make the prediction herself, she entered her final prediction on the next screen. If she delegated instead, this screen was skipped.

We then asked Player A to report her beliefs about Player B’s punishment in each of the four scenarios resulting from the combination of delegation decision (yes/no) and prediction outcome (high/low payoff for Player B). We elicited beliefs on the same scale as actual punishment (£0–£2 in £0.10 increments) and presented the four scenarios in block-wise random order. We chose not to incentivize belief elicitation for three reasons. First, only two of the four scenarios remain payoff-relevant after the delegation decision, making the others purely hypothetical. Second, an incentivized mechanism could induce hedging. Third, an incentive-compatible scoring rule across four scenarios would have substantially increased task complexity. Because we are nonetheless interested in comparing punishment beliefs to actual punishment, we included an attention check on the belief-elicitation screen. In the *No-Punishment* condition, where no punishment was possible, we elicited beliefs about hypothetical punishment to keep the elicitation identical across conditions.

We then asked Player A to assess her own performance in the first ten rounds by assigning a probability mass (summing to 100%) to each possible score from 0 to 10.¹² We elicited this belief after the delegation decision, and after Player A had been told that the algorithm would perform equally well. While learning about algorithm performance may lead her to revise her beliefs about her own performance, it is precisely this post-update belief that is the analytically relevant input for any subsequent behavior. We also asked Player A to estimate the distribution of scores in the population, assigning to each score from 0 to 10 the share of participants she expected to have achieved it. Combined with her belief about her own score, this locates where she places herself in the distribution and yields a measure of perceived ability relative to other participants. One score was drawn at random at the end of the experiment, and Player A earned £0.30 if her estimate for it fell within five percentage points of the realized population share.

Finally, we elicited risk preferences using an incentivized multiple price list in the style of Holt and Laury (2002) and administered a short questionnaire on socio-demographic characteristics and technology affinity, before presenting Player A with her preliminary payoff.

¹²The elicitation was incentivized. Player A earned an additional £0.30 with the probability she had assigned to her actual score. While this linear scoring rule is not proper as its payoff-maximizing report places all mass on the single most likely score, irrespective of risk attitude, in practice most subjects spread probability mass across several scores.

Player B

After providing informed consent, Player B learned that he had been randomly matched with a Player A, and we introduced him to the task Player A had completed. He learned that his own payoff would be determined by the outcome of Player A’s final prediction - £5 if it fell within 10 lbs of the true weight and £1 otherwise - while Player A’s payoff depended solely on her performance in the initial ten rounds. Crucially, we also told Player B that Player A had the option to delegate the final prediction to an algorithm that performs as well as Player A had performed in the earlier rounds, and that Player A was aware of this, making it common knowledge.

In the *Punishment* condition, Player B could choose to reduce Player A’s payoff by up to £2 in £0.10 increments. We focus on punishment rather than reward because the responsibility-attribution literature has predominantly operationalized blame through punishment (Bartling and Fischbacher, 2012; Coffman, 2011; Steffel et al., 2016). Imposing strictly positive punishment required Player B to forgo £0.10 of his own earnings, ensuring that punishment was costly to the punisher. We elicited punishment decisions for all four combinations of delegation (yes/no) and outcome (high/low payoff for Player B), implementing only the decision matching the realized combination. Because the punishment decision is the central outcome variable for Player B, we included an attention check on the punishment-elicitation screen. In the *No-Punishment* condition, where actual punishment was not possible, we elicited hypothetical punishment decisions on the same scale to keep the experimental flow comparable across conditions. Player B’s perceptions of task difficulty, his risk preferences, and a short questionnaire followed, before we presented him with his final payoff.

After Player B’s decisions were recorded, we calculated Player A’s final payoff and paid all participants on the day of participation.

Hypotheses and identification

Our design tests two main hypotheses, both grounded in the literature on responsibility avoidance through delegation to human intermediaries (Bartling and Fischbacher, 2012; Coffman, 2011; Oexl and Grossman, 2013; Hamman et al., 2010; Hill, 2015; Steffel et al., 2016) and on responsibility attributions to algorithmic decision-makers (Feier et al., 2022; Kirchkamp and Strobel, 2019; Gogoll and Uhl, 2018). These strands of the literature mostly focus on what Gogoll and Uhl (2018) have coined the *moral* domain. Although the underlying prediction is not itself moral, Player A’s choice imposes monetary consequences on a matched recipient, and a poor prediction causes real monetary harm. In this respect, our setting shares the defining feature of the settings this literature studies. [Kirchkamp & Strobel is about sharing a decision, not delegating it]

H1 (Delegation): The introduction of potential punishment increases the likelihood that

Player A delegates the final prediction to the algorithm.

H1 is grounded in empirical evidence that responsibility avoidance shapes delegation decisions involving human intermediaries. [Bartling and Fischbacher \(2012\)](#) also show that subjects are more likely to delegate morally consequential decisions even to a die roll, an early form of delegation to a non-intentional agent, suggesting that responsibility avoidance persists when the agent lacks intentionality. Whether the same logic extends to *algorithmic* delegates, to decisions under genuine uncertainty, and to settings in which the outcome falls entirely on the recipient is less explored. [Feier et al. \(2022\)](#) report that delegation rates are similar for human and algorithmic intermediaries, suggesting that the kind of intermediary may matter less than the mere availability of one. Their design, however, does not directly compare delegation rates across punishment regimes, leaving open the question H1 addresses. H1 is identified by the between-subject comparison of Player A’s delegation rate across the *Punishment* and *No-Punishment* conditions, which exploits the punishment manipulation as the only between-subject difference and holds fixed every other feature of the decision environment.

H2 (Punishment): Conditional on a low-payoff outcome, Player B punishes Player A less when she delegated the final prediction to the algorithm than when she made it herself.

H2 mirrors H1 on the punishment side. [Bartling and Fischbacher \(2012\)](#) find that delegation to a randomization device still allows decision-makers to avoid some punishment, although less so than delegation to another human. [Feier et al. \(2022\)](#) find that decision-makers are rewarded less when a negative outcome is directly attributable to their own action, compared to when it results from an algorithm to which they delegated. Our study differs from theirs in three respects. First, the task differs. Second, the roles of Player A and Player B are separated across disjoint subject pools. Third, their design does not control for relative performance at the individual level. Under their calibration, the delegation decision carries a strong signal about Player A’s beliefs concerning her own ability relative to the algorithm. A failure to delegate may then be punished as overconfidence rather than for assuming responsibility, confounding inference on responsibility-shifting through pure intermediation with an inference about ability. H2 is identified by the within-subject comparison across the four (delegation, outcome) scenarios of Player B’s strategy-method punishment, which isolates the effect of delegation on punishment, holding fixed both the realized consequences for Player B and all between-subject heterogeneity in Player B’s punishment proclivity.

4 Results

Data collection took place between February and June 2023 through the British online platform Prolific, with participation restricted to UK residents. A total of 322 subjects were recruited

(161 assigned to Player A, 161 to Player B).¹³ On average, Player A spent 12 minutes completing the experiment and earned £3.63, while Player B spent approximately 8 minutes and earned £2.40. Treatment randomization was effective, as demonstrated by the balance across observable characteristics (Appendix A.1, Table 4).

In the following, we first test the two main hypotheses derived from our preregistered research questions. The subsequent analyses serve to understand the mechanism behind these results. They are exploratory in nature, though several of them were outlined as secondary analyses in the pre-registration.¹⁴

Player A – Delegation behavior

We begin by examining the delegation behavior of Player A. In the *Punishment* condition, where Player B can punish, 42.5% of Player As delegated the final prediction to the algorithm. In the *No-Punishment* condition, where punishment is impossible, 59.3% delegated. The introduction of potential punishment thus significantly *reduces* delegation by 16.8 percentage points (Pearson Chi-squared, $p = 0.033$), opposing the direction posited by H1.¹⁵ Figure 2 illustrates the difference. Delegation shares around 50% are consistent with subjects, on average, finding the claim of equal relative performance between themselves and the algorithm credible. Their behavior reflects neither systematic preference for one’s own prediction nor systematic deference to the algorithm.

Result 1. *The possibility of punishment significantly reduces the likelihood that Player A delegates the final prediction to the algorithm.*

We next examine the robustness of the observed treatment difference in the delegation share and assess whether the treatment difference in delegation survives conditioning on socio-demographic characteristics and on Player A’s actual and perceived task performance. Table 1 reports Probit estimates of the delegation decision.

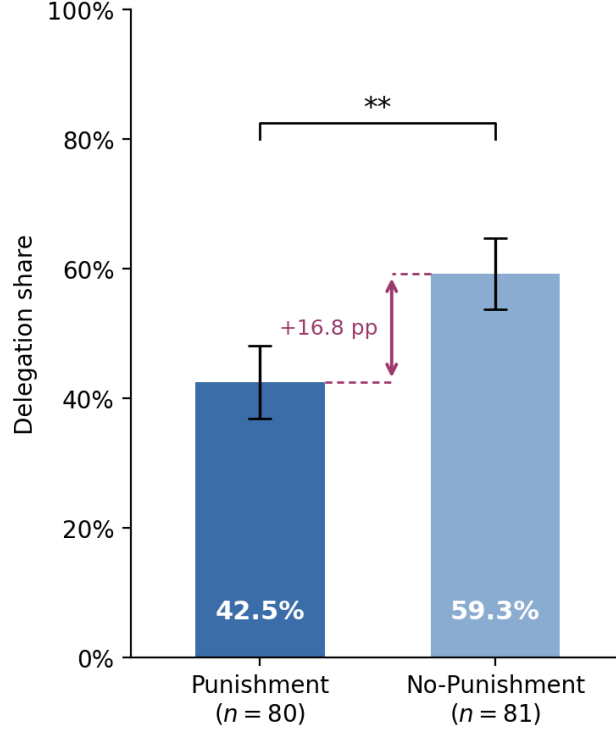
Column (1) reports the unconditional treatment effect. Column (2) adds socio-demographic and other background characteristics. Columns (3) and (4) add Player A’s actual performance

¹³We preregistered the collection of data from 320 participants (80 subjects per role per treatment). Due to attrition and subsequent re-matching we recruited an extra pair in the *No-Punishment* condition.

¹⁴Unless stated otherwise, all tests are two-sided. We describe results as significant if $p < 0.05$ and as weakly significant if $0.05 \leq p < 0.10$. Because the delegation decision is a binary outcome compared between conditions, delegation shares are tested using Pearson Chi-squared tests, with Fisher’s exact tests yielding the same conclusions throughout. Because punishment is elicited within-subject through the strategy method, punishment comparisons use paired t -tests, with Wilcoxon signed-rank tests as non-parametric counterparts. Regressions report robust (HC1) standard errors. The pre-registration and any deviations from the preregistered analyses are described in Appendix C.

¹⁵A within-subject corroborator points in the same direction. Player As in the *No-Punishment* condition were asked, after their actual delegation decision, the hypothetical question of whether they would have delegated had punishment been possible. Among the 81 respondents, 12 would have stopped delegating under hypothetical punishment (25% of delegators) while only 5 would have started (15.2% of non-delegators), constituting weakly significant switching behavior towards making the prediction oneself with punishment possible (within-subject McNemar’s test $p = 0.090$), and yielding a hypothetical delegation rate of 50.6%.

Figure 2: Delegation shares by condition



Note: Bars show the share of Player As who delegated the final prediction to the algorithm in each condition, with whiskers denoting one standard error of the mean. The treatment difference is significant at the 5% level under both a Pearson Chi-squared test ($p = 0.033$) and a Fisher’s exact test ($p = 0.041$); ** $p < 0.05$.

(number of correct predictions in the first ten rounds) and her weighted-average belief about her own performance, respectively. Column (5) restricts the sample to subjects who passed the attention check.¹⁶ The treatment effect is robust across all five specifications. In terms of average marginal effects, introducing the punishment possibility reduces the probability of delegation by between 16.8 and 19.4 percentage points (all $p \leq 0.031$).¹⁷ Neither actual nor perceived performance is significantly associated with delegation in the pooled sample, although actual performance enters negatively and is weakly significant under the attention restriction ($p = 0.068$). We return to this pattern in discussing the mechanisms.

Player B – Punishment behavior

We now turn to Player B’s punishment behavior in the *Punishment* condition, in which punishment was payoff-relevant. Punishment is substantial in all four (delegation, outcome) scenarios of the strategy method, with average deductions between £0.30 and £0.55 (Figure 3). Punishment responds strongly to the outcome. Player Bs punish bad outcomes significantly more

¹⁶While the attention check was included on another screen, it may still correlate with who took the delegation decision seriously.

¹⁷Including a control for the random order in which the two delegation options appeared on screen leaves the treatment coefficient essentially unchanged (Punishment coefficient -0.48 ($p = 0.019$), with the “algorithm first” order indicator at 0.27 ($p = 0.21$), starting from the specification in Column (3)).

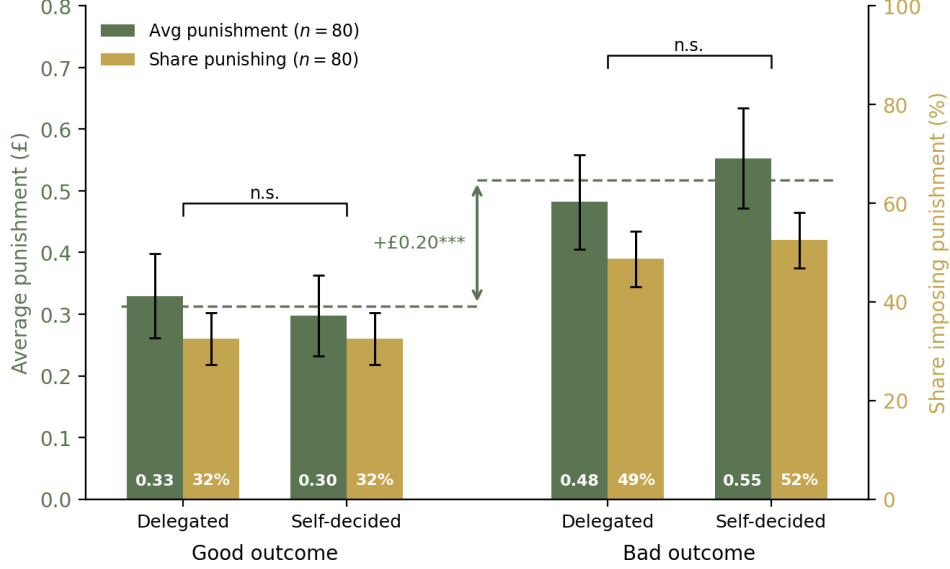
Table 1: Determinants of the delegation decision

	Dependent variable: <i>Delegation</i>				
	<i>Full sample</i>				<i>Passers</i>
	(1)	(2)	(3)	(4)	(5)
Punishment indicator	−0.423** (0.199)	−0.435** (0.204)	−0.472** (0.206)	−0.448** (0.205)	−0.517** (0.230)
Task performance			−0.119 (0.079)		−0.158* (0.086)
Perceived performance				−0.072 (0.056)	
Age		−0.002 (0.009)	−0.002 (0.009)	−0.002 (0.009)	0.002 (0.010)
Female		0.175 (0.227)	0.222 (0.228)	0.212 (0.229)	0.407 (0.254)
Socio-economic status		−0.038 (0.071)	−0.035 (0.072)	−0.031 (0.071)	−0.082 (0.079)
Went to uni		0.036 (0.227)	0.059 (0.225)	0.025 (0.227)	0.124 (0.247)
Technology score		−0.069 (0.092)	−0.048 (0.095)	−0.062 (0.092)	0.014 (0.107)
Leadership position		0.404* (0.225)	0.422* (0.226)	0.399* (0.225)	0.253 (0.253)
Constant	0.234* (0.141)	0.452 (0.545)	0.714 (0.584)	0.716 (0.593)	0.651 (0.656)
AME of Punishment (pp)	−16.8** ($p = 0.031$)	−16.8** ($p = 0.030$)	−18.0** ($p = 0.019$)	−17.2** ($p = 0.026$)	−19.4** ($p = 0.021$)
N	161	159	159	159	130
Pseudo R^2	0.020	0.039	0.049	0.047	0.064

Note: Coefficients from a Probit regression of the delegation decision on the Punishment indicator and a set of controls. Robust (HC1) standard errors in parentheses. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$ (two-sided). Due to missing socio-demographic data, two subjects are dropped in Columns (2)–(5). The *AME of Punishment* reports the average change in the predicted probability of delegation from introducing the punishment possibility, in percentage points.

than good outcomes, by £0.20 on average (paired t -test, $p < 0.001$), indicating that measured punishment carries real informational content rather than being random. The substantial punishment after *good* outcomes need not reflect noise either. Player Bs can hold views about how the final prediction should have been made, independently of the realized outcome, and can punish accordingly.¹⁸

Figure 3: Player B punishment by delegation and outcome scenario



Note: For each (delegation, outcome) scenario, the green bars report the average punishment chosen by Player B (left y -axis, in pounds) and the gold bars the share of Player Bs imposing non-zero punishment (right y -axis, in percent), in the *Punishment* condition. Whiskers are ± 1 standard error of the mean. The brackets mark the within-outcome comparisons of average punishment between delegated and self-decided predictions (paired t -tests, $p = 0.53$ for good and $p = 0.13$ for bad outcomes). The dashed lines mark the mean of average punishment within each outcome group. The subsample of attention-check passers is shown in Appendix Figure 6. Hypothetical punishment in the *No-Punishment* condition is reported in Appendix Figure 8.

Holding the outcome fixed, Player B punishes delegated and self-made decisions no differently. After a good outcome, average punishment is £0.33 for delegated and £0.30 for self-made decisions (paired t -test, $p = 0.53$). After a bad outcome, £0.48 versus £0.55. The punishment gap for bad outcomes thus runs in the direction predicted by H2 but is small and statistically insignificant (£0.07; paired t -test $p = 0.13$; Wilcoxon signed-rank $p = 0.54$).

Result 2. *Conditional on a low-payoff outcome, Player B does not punish delegated decisions significantly less than self-made ones.*

Regression analysis in Table 2 confirms this pattern. Punishment loads overwhelmingly on the outcome, while the effect of delegation for good outcomes is essentially zero. The *Delegated* $\times$ *Bad outcome* interaction, which captures any insulation from punishment after bad outcomes, amounts to -0.103 and is weakly significant ($p = 0.08$).¹⁹ The conclusion is un-

¹⁸For example, Coffman (2011) likewise documents non-trivial punishment in settings where the decision-maker's action could not have been improved upon.

¹⁹In the regression, because punishment for the four scenarios is elicited within-subject, any stable characteristic of Player B is differenced out and cannot, by construction, confound the within-subject comparison. Thus,

changed when we restrict the sample to subjects who passed the attention check, which was included on the punishment elicitation screen (Table 2, Column (2)).

[It is unclear whether we need the diff-in-diff table.]

Table 2: Determinants of Player B’s chosen punishment

	Dependent variable: <i>Punishment</i> (£)	
	<i>Full sample</i>	<i>Passers</i>
	(1)	(2)
Delegated	0.032 (0.052)	0.001 (0.054)
Bad outcome	0.256*** (0.071)	0.260*** (0.080)
Delegated × Bad outcome	−0.103* (0.058)	−0.078 (0.052)
Player B belief about Player A perf.	−0.020 (0.032)	0.001 (0.037)
Age	0.002 (0.005)	−0.001 (0.006)
Female	−0.100 (0.128)	−0.064 (0.140)
Socio-economic status	0.083* (0.047)	0.072 (0.049)
Went to uni	−0.196 (0.140)	−0.098 (0.142)
Technology score	0.073 (0.053)	0.030 (0.059)
Leadership position	0.083 (0.142)	0.061 (0.164)
Constant	−0.077 (0.328)	0.020 (0.374)
Observations	320	268
Subjects	80	67
Adj. R^2	0.077	0.030

Note: OLS regression of Player B’s chosen punishment levels. The sample is stacked over the four (delegation, outcome) scenarios of the strategy method (4 observations per Player B). Column (1) uses the full Punishment-condition sample. Column (2) restricts to Player Bs who passed the attention check on the punishment-elicitation screen. Standard errors clustered at the Player B level.

The same pattern holds on the extensive margin.²⁰ For each outcome, the share of Player Bs imposing *any* punishment is indistinguishable between delegated and self-made decisions (Figure 3, right axis). Punishment is also far from universal. Nearly half of Player Bs (45.0%) impose no punishment in any of the four scenarios. Within each scenario, average punishment therefore reflects a punishing minority rather than light, diffuse sanctioning.

In sum, Player B punishes delegated decisions neither less harshly nor less frequently than self-made ones. In our setting, Player A can therefore not escape punishment by delegating the final prediction to the algorithm, but she also pays no punishment premium for doing so. Punishment responds almost entirely to the realized outcome.²¹ This result also bears on *Result*

the control variables merely capture the between-subject association with Player B’s *average* punishment level. We find no striking relationship between Player B’s average punishment level and either their socio-demographic characteristics or their beliefs about Player A’s *absolute* performance.

²⁰The corresponding regression results can be found in Appendix Table 5.

²¹Hypothetical punishment choices elicited from Player Bs in the *No-Punishment* condition corroborate this

1 as any treatment difference in delegation frequencies driven by punishment avoidance would be inconsistent with Player B’s observed punishment behavior.

Mechanisms

Result 1 does not merely fail to support *Hypothesis 1*, but the effect is statistically significant in the opposite direction. The possibility of punishment discourages, rather than encourages, delegation to the algorithm. Thus, the prospect of punishment changes Player A’s delegation decision, even though we detect no punishment penalty for delegating. In the remainder of this section, we examine the data more closely and weigh candidate explanations with the goal of reconciling the two main results.

Departures from canonical settings. *Hypothesis 1* drew on evidence from settings that differ from ours in ways that plausibly matter. In the modified dictator games of [Bartling and Fischbacher \(2012\)](#); [Coffman \(2011\)](#); [Hamman et al. \(2010\)](#); [Oexl and Grossman \(2013\)](#), the decision-maker chooses between a fair and an unfair allocation. Her intent is therefore transparent, the chosen allocation is implemented with certainty, and the unfair option benefits her directly at the recipient’s expense. Punishment in these settings sanctions a deliberate, self-serving act, and delegation is attractive precisely because it blurs the link between the principal and that act. Our prediction task departs from these settings. The outcome is uncertain, so a bad outcome may reflect bad luck rather than bad intent, and neither option affects Player A’s own monetary payoff, leaving no self-serving choice to police. With little intent worth sanctioning, any intention-based punishment motive may lose its bite, and with it does the value to Player A of placing an algorithm between herself and the outcome. We nevertheless chose this task deliberately since prediction under genuine uncertainty, by a decision-maker with no direct material stake in the affected party’s outcome, is the environment in which real-world algorithmic delegation typically operates. These departures, however, may explain the absence of the predicted effect, not its reversal. A reversal requires a motive that weighs against delegation when punishment is possible.

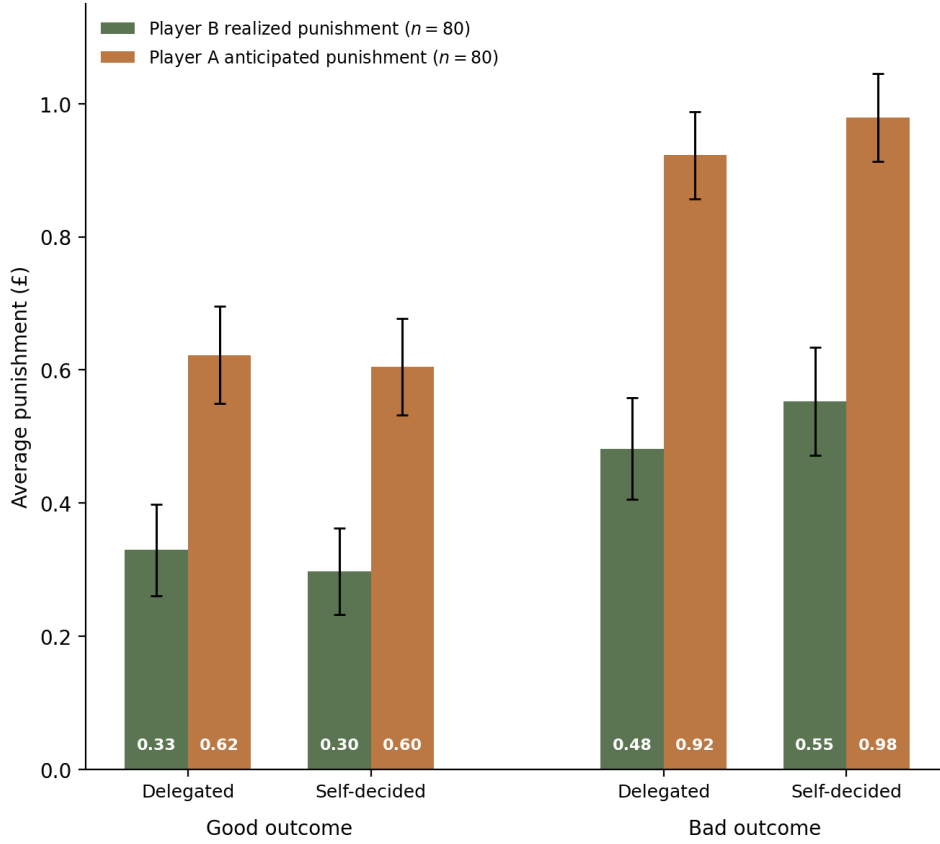
Anticipated differential punishment. The most natural candidate is anticipation of differential punishment. Player A may expect a delegated decision to be punished more harshly than a self-made one, and delegate less in response. Such an expectation would, for example, be consistent with previous evidence that observers view the delegation of morally consequential decisions to machines relatively critically (e.g., [Gogoll and Uhl, 2018](#); [Bigman and Gray, 2018](#); [Jauernig et al., 2022](#); [Bonnefon et al., 2024](#)).

[Inclusion of shirking here?]

conclusion (Appendix Figure 8). If anything, these Player Bs state that they would punish a delegated decision *more* than a self-made one after a bad outcome (£0.53 versus £0.43; paired *t*-test, $p = 0.069$).

After the delegation decision but before learning the outcome, we elicited Player A’s beliefs about Player B’s punishment in each of the four (delegation, outcome) scenarios. Anticipated punishment can affect behavior only where punishment is possible. We therefore focus the analysis of punishment beliefs on the *Punishment* condition. Two patterns emerge (Figure 4). First, Player As, on average, substantially overestimate punishment in every scenario. Second, Player As expect essentially the same punishment for delegated and self-made decisions, conditional on the outcome, correctly anticipating the pattern in Player B’s actual punishment behavior (Result 2).²² Thus, elicited beliefs provide no support for a punishment anticipation mechanism for Result 1.²³

Figure 4: Realized vs. anticipated punishment, by scenario



Note: Player B’s realized punishment (green bars) and Player A’s anticipated punishment (terracotta bars), in pounds, across the four (delegation, outcome) scenarios in the *Punishment* condition (full sample). Whiskers denote ± 1 standard error of the mean. The subsample of attention-check passers is shown in Appendix Figure 7.

Player As, however, are not unanimous in the punishment penalty they expect. We therefore ask whether these individual differences predict delegation. Table 3 reports Probit regressions of the delegation decision on Player A’s belief difference in expected punishment (*no delegation* – *delegation*), so that higher values correspond to a larger expected punishment penalty for deciding herself. All specifications are restricted to the *Punishment* condition. Columns (1)–(3)

²²The expected punishment penalty for delegation is statistically indistinguishable from zero in both outcome states, and a within-subject test finds no asymmetry between the two.

²³The same holds for the fully hypothetical beliefs elicited in the *No-Punishment* condition, which closely track those in the *Punishment* condition (Appendix Figure 8). The treatment manipulation itself therefore appears not to have changed what subjects expected punishment to be.

use the simple average of the belief differences for good and bad outcomes, Column (4) replaces the simple average with a weighted average, using Player A's elicited probability of producing the high payoff as the weight, and Column (5) enters the two differences separately.

Table 3: Delegation and beliefs about punishment

	Dependent variable: <i>Delegation</i>				
	<i>Full Punishment</i>		<i>Punishment, passers</i>		
	(1)	(2)	(3)	(4)	(5)
Belief difference, average ($\bar{\Delta}$)	0.346 (0.407)	0.398 (0.448)	1.661** (0.700)		
Belief difference, weighted by $\Pr(\text{good})$ (Δ^w)				1.992** (0.794)	
Belief difference, bad outcome (Δ^{bad})					0.590 (0.424)
Belief difference, good outcome (Δ^{good})					0.961** (0.412)
Task performance		-0.208* (0.111)	-0.362*** (0.136)	-0.342** (0.142)	-0.381*** (0.138)
Age		0.023* (0.012)	0.039** (0.016)	0.040** (0.016)	0.039** (0.016)
Female		0.152 (0.350)	0.950** (0.427)	1.026** (0.406)	0.845* (0.444)
Socio-economic status		-0.049 (0.111)	-0.006 (0.131)	-0.055 (0.133)	0.001 (0.126)
Went to uni		0.071 (0.363)	-0.278 (0.412)	-0.169 (0.408)	-0.275 (0.409)
Technology score		-0.061 (0.138)	0.048 (0.170)	0.072 (0.173)	0.033 (0.167)
Leadership position		0.302 (0.312)	-0.301 (0.408)	-0.279 (0.408)	-0.245 (0.411)
Constant	-0.198 (0.142)	-0.345 (0.826)	-1.228 (1.155)	-1.281 (1.147)	-1.115 (1.165)
AME of belief measure (pp per £0.10)	1.3 ($p = 0.388$)	1.4 ($p = 0.370$)	4.7** ($p = 0.011$)	5.5*** ($p = 0.006$)	
AME: good-outcome difference (pp per £0.10)					2.7** ($p = 0.010$)
AME: bad-outcome difference (pp per £0.10)					1.7 ($p = 0.165$)
Controls	×	✓	✓	✓	✓
Passers only	×	×	✓	✓	✓
N	80	78	60	60	60
Pseudo R^2	0.007	0.098	0.248	0.263	0.252

Note: Probit estimates with robust (HC1) standard errors in parentheses. Sample restricted to the *Punishment* condition throughout. Belief differences are within-subject (*no delegation*) – (*delegation*) in expected punishment, so positive coefficients indicate that subjects who expect to be punished more harshly for not delegating (relative to delegating) are more likely to delegate. Columns (1)–(3) use the simple average of the good-outcome and the bad-outcome belief difference. Column (4) instead weights the good-outcome difference by Player A's elicited probability of producing the high payoff and the bad-outcome difference by the complementary probability. Column (5) enters the two differences separately. The *AME* rows report the average marginal effect of the belief measure entered in the respective column on the probability of delegation, in percentage points per £0.10 increase in the corresponding belief difference.

Expected differential punishment does not predict delegation in the full sample (Columns (1)–(2)). Among subjects who passed the attention check on the belief-elicitation screen, the relationship sharpens. A £0.10 larger expected punishment penalty for deciding herself is associated with a 4.7 percentage-point higher probability of delegation (Column (3), $p = 0.011$),

and with 5.5 percentage points under the weighted specification (Column (4), $p = 0.006$).²⁴

Taken together, the belief evidence does not support anticipated differential punishment as the mechanism behind Result 1. For anticipation of punishment to generate a 17 percentage-point delegation gap between conditions, Player As would need to expect systematically harsher punishment for delegated decisions. The measured beliefs show no such expectation, and such an expectation would not have been correct, as Player B’s realized punishment exhibits no delegation penalty either (Result 2). What the belief evidence does show is heterogeneity. Within the *Punishment* condition, Player As whose beliefs embody a larger penalty for deciding themselves are more likely to delegate. Anticipated punishment is thus a dimension along which subjects differ, but it provides no aggregate push toward less delegation under punishment.²⁵ We nonetheless read the belief evidence with caution. First, hypotheticality may introduce noise as only two of the four scenarios could still materialize after Player A’s delegation decision in the *Punishment* condition.²⁶ Second, the elicitation of beliefs was unincentivized and took place after the delegation decision, so reports may reflect post-hoc rationalization rather than the beliefs that drove the choice.

Result 3. *On average, Player A expects no punishment penalty for delegation, mirroring Player B’s actual punishment behavior. Within the Punishment condition, Player As who expect a larger penalty for deciding themselves are more likely to delegate.*

Sample composition. Result 1 is not an artifact of sample composition either. Treatment randomization produced balance across socio-demographic and other background characteristics (Appendix A.1, Table 4). The same holds for performance. Average performance in the first ten rounds is balanced across conditions (2.76 vs. 3.09 correct predictions out of ten in the *Punishment* and *No-Punishment* conditions, respectively; Figure 5).²⁷ Player As in both conditions also exhibit similar overconfidence about their own performance (4.65 vs. 4.87, respectively).²⁸ Finally, the treatment effect is robust to controlling for actual and perceived performance (Table 1, Columns (3)–(4)).

²⁴In the decomposition of Column (5), the good-outcome belief difference carries the relationship (2.7 percentage points per £0.10, $p = 0.010$), while the bad-outcome difference is positive but noisier (1.7 percentage points, $p = 0.165$). The two differences are nearly orthogonal in this sample [How does this relate to the earlier statement that there is no within-subject asymmetry?], so we read their coefficients as additive partial slopes rather than competitors for the same variance. Under expected-utility reasoning, however, only their probability-weighted sum should enter the decision rule. We therefore treat Column (5) merely as a descriptive decomposition. [Does this not only hold under risk neutrality or other symmetric motives?]

²⁵A mean expected penalty of zero need not rule out an effect on delegation if Player As react more strongly when they expect a penalty for delegating than when they expect one for deciding themselves. We find no such asymmetry.

²⁶In the *No-Punishment* condition all four scenarios were purely hypothetical.

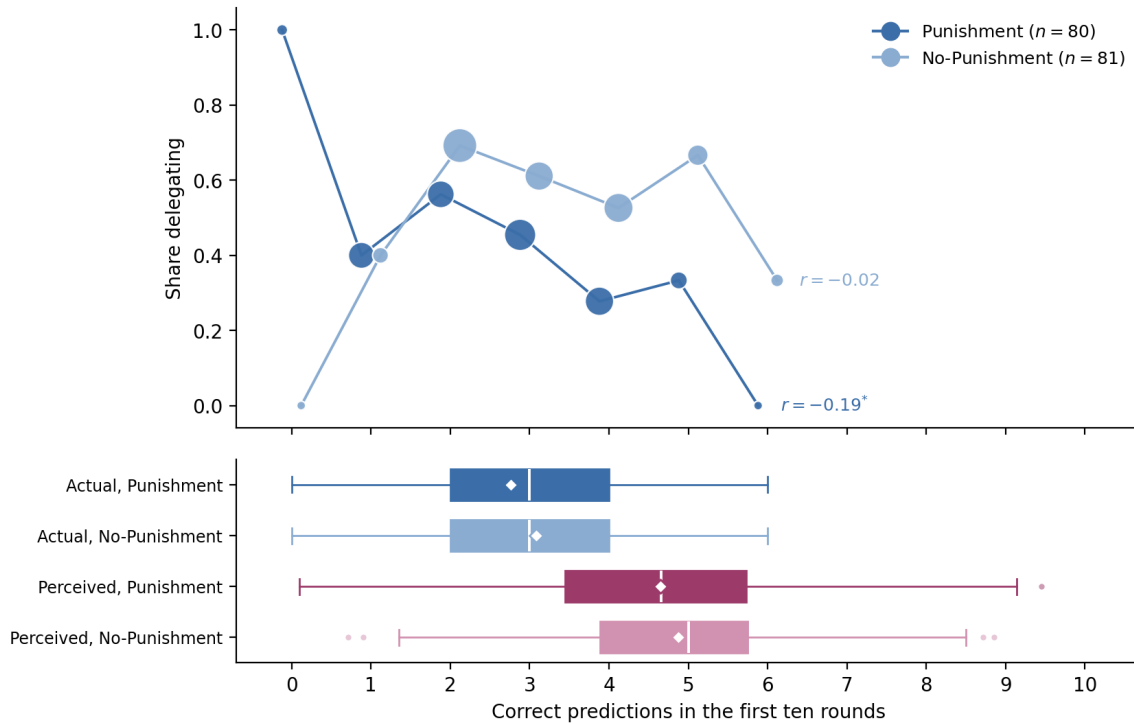
²⁷To the extent this small raw difference matters, it works against Result 1, because Player A in the *Punishment* condition performs slightly worse, and as we will see, it is worse performers who tend to delegate more, so imbalance would attenuate rather than generate the treatment difference.

²⁸Confidence is measured by the weighted-average belief about own performance. This overconfidence may partly reflect the information that the algorithm performs equally well.

Performance and delegation. What differs across conditions is who delegates. Task performance is negatively associated with the decision to delegate across conditions (Table 1, Column (5)), and this relationship is considerably stronger in the *Punishment* condition (Table 3) than in the *No-Punishment* condition (Figure 5). For perceived performance, a directionally similar but much weaker pattern emerges in the *Punishment* condition.²⁹ Thus, actual performance separates delegators from non-delegators under the prospect of punishment and the delegation gap between conditions therefore widens with performance. Player As who delegated in the *Punishment* condition scored on average 2.47 out of 10 in the first ten rounds, compared with 3.06 for delegators in the *No-Punishment* condition (t -test, $p = 0.039$). It is precisely the better performers who, when punishment is on the table, keep the decision for themselves.

Result 4. *In the Punishment condition, better-performing Player As are less likely to delegate. The additional delegation in the No-Punishment condition comes disproportionately from better performers.*

Figure 5: Delegation, performance, and confidence by condition



Note: The top panel shows the delegation share of Player A at each score, by treatment. Dot size is proportional to the number of subjects at that score. The bottom panel shows the distributions of actual and perceived (the weighted-average belief about own performance) scores, by condition. Boxes span the interquartile range, the white line marks the median, and the white diamond the mean.

Our design rules out the natural explanation for this relationship, namely that worse performers delegate because they believe the algorithm outperforms them. In [Feier et al. \(2022\)](#), the algorithm's performance distribution is matched to the distribution of human performance at

²⁹Player A's weighted-average belief about her own performance would enter the regressions with a small negative coefficient but is never statistically significant.

the population level. A high-performing subject who recognizes her position in that distribution therefore rationally expects to outperform the algorithm, and a low-performing subject rationally expects the reverse. Delegation that declines with performance is then exactly what expected-payoff maximization predicts. In our experiment, by contrast, performance is equalized between the algorithm and each individual subject, and this equality of performance is common knowledge. The pattern in the *Punishment* condition must therefore reflect something other than beliefs about relative ability.³⁰

One reading of the relationship between performance and delegation attributes it to a preference over *process ownership* of the outcome. Player As with a good chance of producing the high payoff may prefer to bring it about themselves rather than to have the algorithm do so, a preference that is strongest for the better performers and so tracks the negative relationship between performance and delegation. A symmetric version of this preference, in which Player As also wish to avoid authoring bad outcomes, would be equally consistent with this relationship but would, if anything, push the predominantly low-performing Player As toward delegation and predict the opposite treatment effect.

A related reading concerns the uncertainty inherent in the task itself. Dietvorst and Bharti (2020) document that the preference for algorithms declines as outcome uncertainty rises. In our setting, outcome uncertainty is arguably greatest when the chance of success is near one half, and because most subjects score below 5 out of 10, better performance moves the success probability toward that point.³¹ A distaste for delegating uncertain decisions would then, like process ownership, depress delegation among better performers.

However, both readings describe a preference that is present whether or not punishment is possible, so each would affect delegation in both conditions alike and cannot on its own explain the difference between them. Either can account for the treatment difference only if the prospect of punishment activates it. For example, even with equal expected punishment, being punished for a decision one gave away may feel different from being punished for a decision one made oneself. Or relinquishing control over an uncertain outcome may trouble Player A only once the outcome carries some consequences for her, which punishment brings about. However, we cannot test either story and offer them only to show that activation by the prospect of punishment is conceivable. Consistent with such activation, the relationship between performance and delegation is markedly stronger under punishment than without it.

³⁰We cannot rule out that some subjects did not fully internalize the individual-level equalization. Two observations mitigate this concern. First, any such misunderstanding should arise in both conditions alike, so it can explain neither why the relationship between performance and delegation is markedly stronger under punishment nor why delegation rates differ between conditions. Second, under this alternative, delegation should decline with perceived own performance, with more confident subjects retaining the decision. In the data, perceived performance is statistically indistinguishable between delegators and non-delegators.

³¹Outcome uncertainty here is distinct from the difficulty of the task, which sets the level of the success probability.

Effort to outperform the algorithm. Lastly, the prospect of punishment could motivate Player A to exert additional effort when making the final prediction for Player B. Although the algorithm’s chance of success is fixed by Player A’s past performance, the final prediction itself has yet to be made. Player A may thus believe that, by trying especially hard, she can outperform the algorithm and thereby give Player B a better chance of the high payoff. To the extent that she expects milder punishment after a good outcome, the prospect of punishment would then push her toward making the prediction herself. This account yields a testable prediction. Relative to the first ten rounds, non-delegators should improve their performance more in the *Punishment* condition than in the *No-Punishment* condition. Similar improvement in both conditions would instead indicate a punishment-independent motive, for instance altruism toward Player B, which cannot explain Result 1.

The data do not support this story. Delegators do not make a final prediction themselves, so effort on the final prediction is observable only for non-delegators. We measure improvement as the difference between a subject’s average absolute prediction error over the first ten rounds and her absolute error on the final prediction. By this measure, non-delegators in the *No-Punishment* condition improve, whereas non-delegators in the *Punishment* condition do not. Because all Player As faced the same images in the first ten rounds and on the final prediction, this contrast cannot reflect variation in image difficulty. Realized outcomes tell the same story. 45.5% of non-delegators in the *No-Punishment* condition earn the high payoff for Player B, compared with 21.7% in the *Punishment* condition, even though the two groups’ performance over the first ten rounds is statistically indistinguishable (3.12 vs. 2.98 correct predictions; t -test, $p = 0.66$).³² If anything, performance on the final prediction is higher where punishment is unavailable, the opposite of what a punishment-motivated effort account predicts.³³ Effort itself remains unobserved, however, and realized outcomes as well as time spent on the decision screen are only imperfect measures of it, so we cannot rule the channel out conclusively.³⁴

Result 5. *The prospect of punishment does not induce additional effort. Non-delegators improve on their performance less in the Punishment condition than in the No-Punishment condition.*

The signaling value of not delegating. Although the prospect of punishment does not appear to induce additional effort (Result 5), the delegation choice itself may communicate effort. By keeping the decision, Player A may want to convey to Player B that she takes

³²Balance among non-delegators across conditions coexists with the selection pattern rather than contradicting it. Non-delegators in the *Punishment* condition are drawn from a slightly weaker pool but are more strongly positively selected within it, so the two forces offset, whereas among delegators they compound.

³³Screen time points in the same direction. In the *Punishment* condition Player As spend less time on the final prediction screen than in the *No-Punishment* condition (medians 16 vs. 22 seconds; Mann-Whitney $p = 0.019$). The same holds within subject, as non-delegators in both conditions spend more time on the final prediction than on an average initial round, but the increase is smaller under punishment (+5.0 vs. +8.2 seconds; Mann-Whitney $p = 0.058$).

³⁴For instance, if the prospect of punishment raised effort while the accompanying pressure depressed performance, additional effort would be present, yet masked.

the task seriously and intends to try hard on his behalf. Delegating, by contrast, could be perceived as shirking a decision that matters to another person. The signal lies in the choice itself rather than in any effort that follows, so Result 5 does not speak against this account. Result 3, however, seems to. The signal has monetary value only if Player A expects delegation, as apparent shirking, to draw harsher punishment, whereas in the *No-Punishment* condition, Player A could shirk without monetary consequences. Result 3 indicates that Player As in the *Punishment* condition, on average, hold no such expectation, which leaves little monetary gain from the signal.³⁵

Two considerations keep this channel open. First, Player A may care about sending this signal, beyond any punishment attached to it. For this concern to explain the treatment difference, however, it would again have to be activated by the prospect of punishment itself, even though the expected amount of punishment does not differ by delegation choice. In our environment, where no self-serving option exists and outcomes are noisy, the delegation choice is the only deliberate act on which Player B can condition punishment, and thus the natural focus of any judgment he forms. Second, as discussed earlier, our punishment belief measure was unincentivized and elicited only after the delegation decision, so it may capture the beliefs behind that decision imperfectly and understate the penalty Player A anticipated for delegating. Either way, a concern about being perceived as shirking would weigh against delegation precisely when punishment is possible.

Taking stock. The reversal in Result 1 is robust, but its mechanism remains unresolved. It is not a composition artifact, since the conditions are balanced in observables, performance, and confidence, and the treatment effect survives the corresponding controls. Belief evidence suggests that anticipated differential punishment cannot explain the reversal either (Result 3). On average, Player A expects no punishment penalty for delegation, mirroring Player B’s realized punishment (Result 2), although Player As who expect a larger penalty for deciding themselves are more likely to delegate. Nor do we find evidence that punishment-induced effort can explain the reversal. Even though actual effort remains unobserved, non-delegators spend less time on the prediction for Player B and improve less under punishment, not more (Result 5). What the data do show is selection. It is the better-performing Player As who retain the decision when punishment is possible (Result 4). Preferences over process ownership or outcome uncertainty are consistent with this selection but operate in both conditions alike. They could explain Result 1 only if accountability itself activated them, a hypothesis our design cannot test. The signaling value of not delegating requires no actual additional effort (Result 5), however, to explain the treatment effect it similarly rests on being activated by the prospect of punishment itself.

³⁵At the individual level, this channel is consistent with the belief heterogeneity documented in Result 3, though a mean expected penalty of zero implies no aggregate push toward less delegation.

5 Conclusion

[On multiple occasions, we say that delegation carries no information about relative ability. Simultaneously, we argue that the delegation decision is a signal about intended effort (and by extension, performance?) in the prediction for Player B. Do we see a contradiction here?]

As algorithms take over decisions that affect other people, who should answer for the results has become the subject of a lively legal, ethical, and policy debate. This paper examines the behavioral side of this debate. We ask whether the prospect of being held responsible shapes the decision to put an algorithm in charge, and whether putting an algorithm in charge shapes the responsibility assignment of those affected by the outcome. We take both questions to a preregistered online experiment in which a decision-maker either makes a payoff-relevant prediction for a recipient herself or delegates it to an algorithm calibrated, subject by subject, to match her own performance. Between subjects, we vary whether the recipient can reduce the decision-maker's earnings. We elicit punishment for every combination of delegation choice and prediction outcome, so we observe directly whether delegation shields the delegator from punishment, conditional on the outcome. Because the two conditions differ only in whether punishment is possible, the comparison of delegation rates identifies the effect of the prospect of punishment on the decision to delegate.

Both answers depart from the canonical account of delegation as a means of responsibility shifting. The prospect of punishment does not push decision-makers toward the algorithm but away from it, a reversal of the prediction imported from settings with human intermediaries (e.g., [Bartling and Fischbacher, 2012](#)). Delegation, in turn, offers no protection against punishment. Recipients punish delegated and self-made predictions alike, conditioning their sanctions almost entirely on the realized outcome rather than on how the prediction was made. The punishment result offers an explanation for why the canonical prediction fails, yet not for its reversal.

We examine several explanations to reconcile these findings. Delegation preferences over task uncertainty or process ownership would explain the treatment difference only if the prospect of punishment itself activated them. Beliefs show no punishment premium attached to delegation. Prediction outcomes and screen times do not indicate punishment-induced additional effort, whereby the decision-maker would retain the decision in the hope of outperforming the algorithm. Merely signaling to exert additional effort by retaining the decision would require no measurable effort at all, but move delegation only if the prospect of punishment gave the signal value. Belief evidence speaks against such a monetary value, leaving only a non-monetary value in which the prospect of punishment makes the recipient's judgment salient in itself. However, both measures are imperfect. The belief elicitation was unincentivized and came only after the delegation decision, so it may capture post-hoc rationalization rather than the beliefs that drove the choice, while actual effort is unobserved, with prediction outcomes and screen times serving only as proxies. The measures may therefore simply fail to detect a decision-maker who fears that delegating will be read as shirking and draw harsher punishment, or who retains the decision to try to outperform the algorithm, a motive that would at least be consistent with

recipients punishing bad outcomes more heavily. These limits leave the mechanism narrowed rather than settled.

Nevertheless, we contribute by providing the first direct test of whether the prospect of punishment motivates delegation to an algorithm, and show that in an environment with inherent task uncertainty and no self-serving choice to police, accountability disciplines algorithm use rather than fueling it. We also show that delegation cannot shift blame when the algorithm performs no better or worse than the decision-maker, with recipients conditioning their sanctions on the outcome rather than on how the prediction was made. More broadly, our findings add a determinant of algorithm use to those documented in the aversion literature, one that operates not through the algorithm or the task but through what delegating may convey about the person who chooses it.

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Appendix

A Additional Tables

A.1 Condition balance — Player A

Table 4: Condition balance — Player A

Variable	<i>Punishment</i>	<i>No-Punishment</i>	<i>p-value</i>
Age	39.026	37.877	0.552
Female	0.487	0.494	1.000
Socio-economic status	5.179	5.321	0.564
Went to uni	0.575	0.556	0.928
Technology score	2.087	2.284	0.322
Leadership position	0.362	0.333	0.824

Note: Mean values by condition for Player A, with p -values from two-sided t -tests for continuous variables and Chi-squared tests for binary variables. *Socio-economic status* is self-reported on a 1–10 scale. *Went to uni* indicates a university degree. *Technology score* is constructed from weekly device usage, programming skills, cryptocurrency knowledge, technology use at work, and similar items. *Leadership position* indicates self-reported management or supervisory experience.

Complementary to the treatment comparisons of individual observable characteristics in Table 4, we can test their joint significance by regressing the treatment indicator on all characteristics in a linear probability model. The observables are jointly uninformative about treatment ($F(6, 152) = 0.36$, $p = 0.90$; $R^2 = 0.01$; $N = 159$). A probit specification yields the same conclusion (likelihood-ratio $\chi^2(6) = 2.27$, $p = 0.89$).

We do not separately report a balance table for Player B. Player B’s punishment behavior is identified within-subject across the four (delegation, outcome) scenarios of the strategy method, so balance across conditions is not strictly required for the punishment analysis.

A.2 Extensive margin of punishment

Table 5 re-estimates the punishment regression on the *extensive* margin, replacing the punishment amount with an indicator for whether Player B imposed any punishment in a given scenario, estimated by probit. The conclusion matches the intensive-margin result in the body. A bad outcome sharply raises the probability of punishment—an average marginal effect of about 18 percentage points ($p < 0.001$)—whereas delegation leaves it essentially unchanged: the average marginal effect of delegation is -1.8 percentage points and statistically insignificant ($p = 0.46$; -2.9 percentage points among attention-check passers, $p = 0.29$). Player A is therefore no less *likely* to be punished when she delegates, just as she is punished no less *harshly*.

Table 5: Extensive margin of punishment — probit for $\Pr(\text{punishment} > 0)$

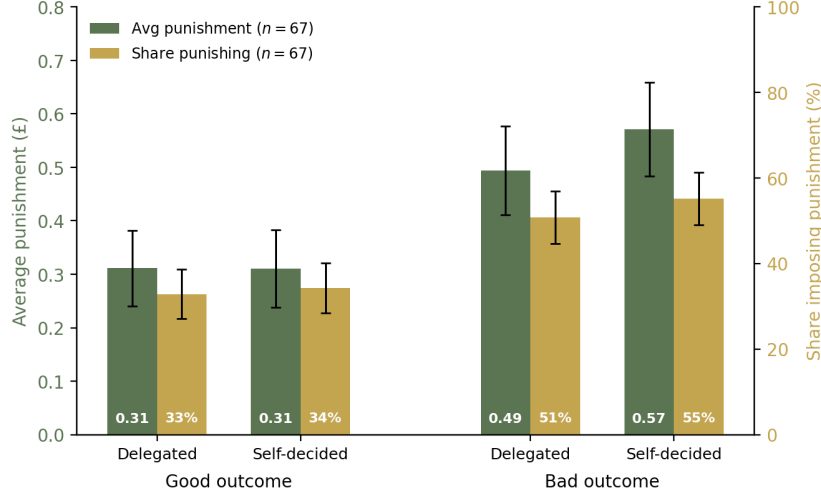
	Dependent variable: <i>Any punishment</i>	
	<i>Full sample</i>	<i>Passers</i>
	(1)	(2)
Delegated	0.002 (0.077)	−0.042 (0.078)
Bad outcome	0.556*** (0.139)	0.563*** (0.153)
Delegated × Bad outcome	−0.099 (0.095)	−0.072 (0.103)
Player B belief about Player A perf.	−0.083 (0.075)	−0.030 (0.082)
Age	−0.006 (0.013)	−0.013 (0.015)
Female	0.115 (0.272)	0.230 (0.291)
Socio-economic status	0.160 (0.126)	0.140 (0.129)
Went to uni	−0.827*** (0.320)	−0.706** (0.349)
Technology score	0.062 (0.118)	−0.023 (0.131)
Leadership position	0.062 (0.293)	−0.071 (0.321)
Constant	−0.382 (0.807)	−0.182 (0.864)
AME of Delegated (pp)	−1.8 ($p = 0.462$)	−2.9 ($p = 0.285$)
AME of Bad outcome (pp)	18.1*** ($p < 0.001$)	19.4*** ($p < 0.001$)
Observations	320	268
Subjects	80	67
Pseudo R^2	0.096	0.081

Note: Probit regression of an indicator for non-zero punishment ($1\{\text{punishment} > 0\}$) on a delegation indicator, a bad-outcome indicator, their interaction, Player B’s belief about Player A’s performance, and socio-demographic controls; the sample is stacked over the four (delegation, outcome) cells of the strategy method (4 observations per Player B). Column (1) uses the full Punishment-condition sample; Column (2) restricts to attention-check passers. Standard errors clustered at the Player B level. The *AME* rows report discrete average marginal effects on the probability of imposing any punishment, in percentage points (for Delegated, the change from $0 \rightarrow 1$ with the interaction moved consistently; for Bad outcome, likewise), computed by counterfactual prediction with p-values from a seeded 400-replication subject-cluster bootstrap. Significance: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$ (two-sided).

B Additional figures

B.1 Punishment among attention-check passers

Figure 6: Player B punishment by delegation and outcome scenario, attention-check passers

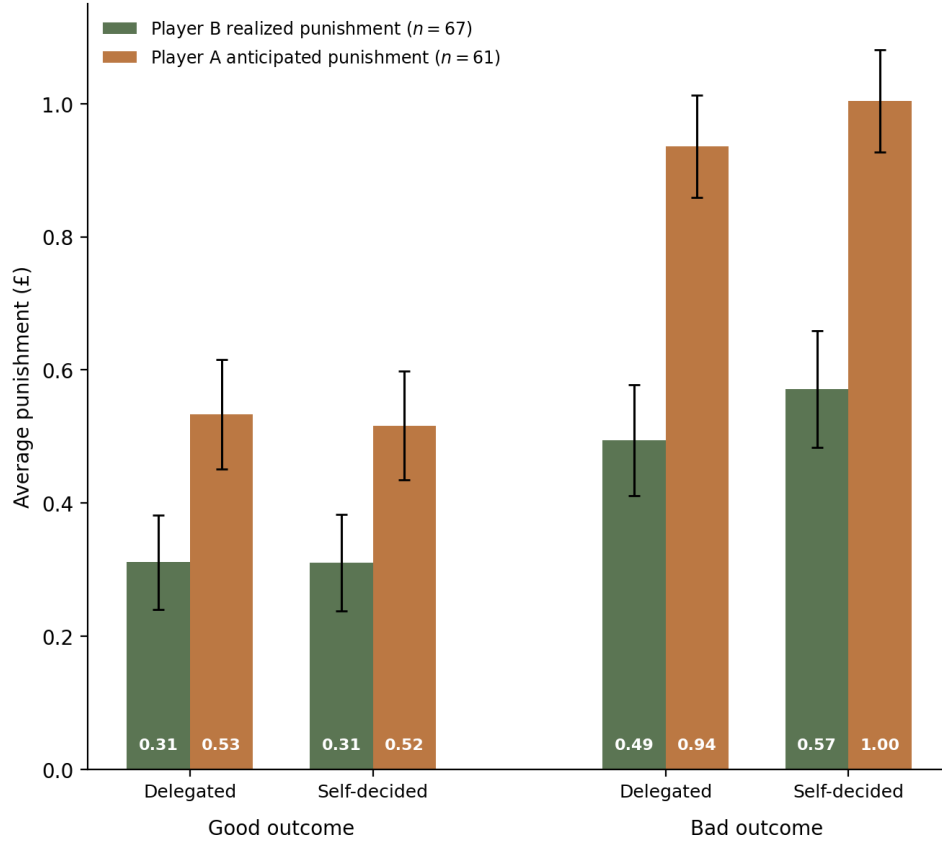


Note: As in Figure 3, but restricted to Player Bs who passed the attention check on the punishment-elicitation screen ($n = 67$). The four (delegation, outcome) scenarios of the strategy method are grouped by outcome. For each scenario, the green bars report the average punishment (left y -axis, in pounds; whiskers are ± 1 standard error of the mean) and the gold bars the share of Player Bs imposing non-zero punishment (right y -axis, in percent; whiskers are ± 1 standard error of the share). Values are printed at the base of each bar. The left-axis scale matches Figure 3.

B.2 Hypothetical punishment

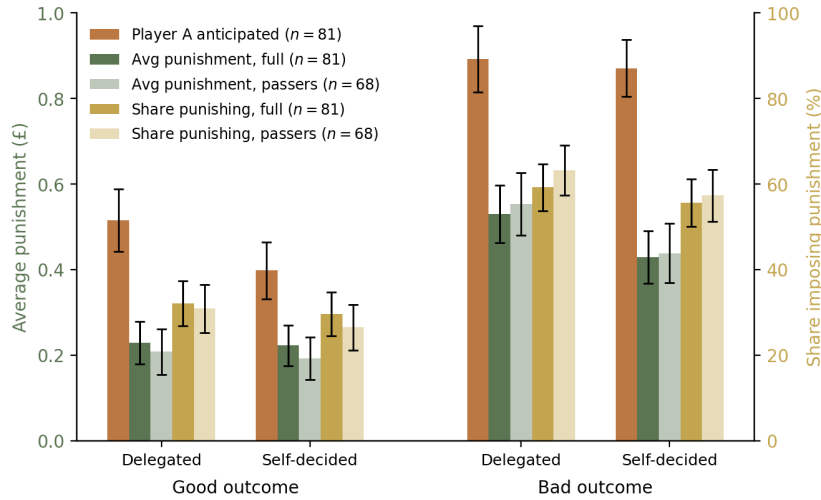
Hypothetical punishment in the *No-Punishment* condition follows the same outcome-driven pattern as actual punishment (Figure 8). After good outcomes, stated punishment is nearly identical for delegated and self-made decisions (£0.23 versus £0.22; paired t -test, $p = 0.91$). After bad outcomes, the comparison reverses relative to the direction predicted by *Hypothesis 2*. Player Bs state that they would punish a delegated decision more than a self-made one (£0.53 versus £0.43; paired t -test, $p = 0.069$), a weakly significant difference. Even in stated terms, delegation thus offers Player A no insulation from punishment. Player A's anticipated punishment, shown alongside, overstates hypothetical punishment in every scenario, mirroring the overestimation of actual punishment in the *Punishment* condition (Figure 4).

Figure 7: Realized vs. anticipated punishment, attention-check passers



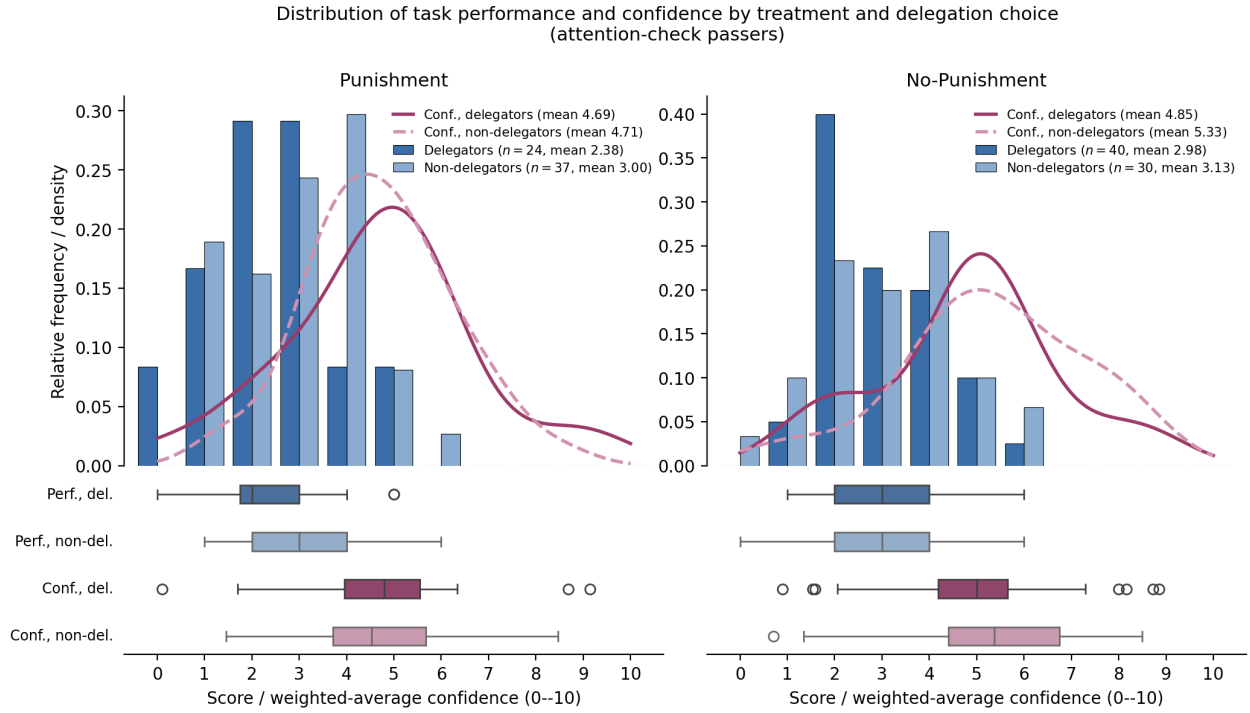
Note: As in Figure 4, but restricted to attention-check passers (Player A $n = 61$, Player B $n = 67$).

Figure 8: Hypothetical punishment by delegation and outcome scenario (*No-Punishment* condition)



Note: As in Figure 3, but for the *No-Punishment* condition (full sample $n = 81$, passers $n = 68$), in which Player Bs answered the same punishment questions but could not impose punishment in practice. The four (delegation, outcome) scenarios are grouped by outcome. For each scenario, the green bars report the average hypothetical punishment (left y -axis, in pounds; whiskers are ± 1 standard error of the mean) and the gold bars the share of Player Bs choosing non-zero punishment (right y -axis, in percent; whiskers are ± 1 standard error of the share); the full sample is shown at full opacity and the attention-check passers at lower opacity. In addition, the terracotta bars report Player A's anticipated punishment in the *No-Punishment* condition (hypothetical, full sample, $n = 81$).

Figure 9: Performance and confidence distributions by treatment and delegation choice



Note: Restricted to attention-check passers. Within each treatment, the top panel shows the share of Player As at each first-ten-rounds score, separately for those who delegated (dark blue) and those who did not (light blue), with the weighted-average performance belief overlaid as a KDE (delegators solid magenta, non-delegators dashed pink). Boxplot strips below the top panel report the same four series. In the *Punishment* condition, actual performance discriminates between delegators and non-delegators (means 2.38 vs. 3.00) while perceived performance does not (means 4.69 vs. 4.71). In the *No-Punishment* condition, neither actual nor perceived performance discriminates (actual: 2.98 vs. 3.13; perceived: 4.85 vs. 5.33). The additional delegators in *No-Punishment* therefore come disproportionately from the middle and upper portion of the actual performance distribution.

Figure 10: Experimental timeline — Player A

To be added.

Figure 11: Experimental timeline — Player B

To be added.

B.3 Performance and confidence by delegation choice

B.4 Experimental timelines

C Pre-registration

The study was pre-registered on AsPredicted under the title “Algorithms and Responsibility” (#133980) on May 30, 2023, after the collection of a pilot sample and before the collection of the main sample. The pre-registration is available at <https://aspredicted.org/hs9az8.pdf> and is reproduced in full on the following page.

The analyses in the paper deviate from this document in three respects. First, we recruited 322 participants instead of the planned 320. Attrition during the experiment required re-matching and produced one additional pair in the *No-Punishment* condition. Second, we compare delegation rates across conditions using a Pearson chi-squared test and Fisher’s exact test rather than the preregistered *t*-test. The delegation decision is binary, so these tests are the appropriate choice, and the conclusion is unchanged under either procedure. Third, the pre-registration specifies Mann-Whitney U and Kolmogorov-Smirnov tests for the punishment comparisons. Both are tests for independent samples, whereas punishment is elicited within-subject via the strategy method. We therefore report the Wilcoxon signed-rank test as the within-subject non-parametric counterpart alongside the parametric tests. Beyond these deviations, the pre-registration left open whether participants who failed the attention checks would be excluded. We report all analyses for the full sample in the main text and repeat them for the subsample of attention-check passers in the appendix.

Algorithms and Responsibility (#133980)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

Pre-registered on:
2023/05/30 16:27 (PT)

1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

We test whether:

RQ 1. punishment for a decision can be avoided if it is delegated to an algorithm (conditional on performance)

RQ 2. a punishment motive enters the delegation decision.

3) Describe the key dependent variable(s) specifying how they will be measured.

The experiment features a delegator and an evaluator. The delegator has to make a decision that is payoff-relevant only to the evaluator. The delegator may decide herself or delegate the decision to an algorithm, which is calibrated to match the delegators performance. Depending on the treatment, the evaluator may punish the delegator conditional on the outcome of the decision and the delegation decision (strategy method).

For RQ 1, we will look at evaluator punishment behavior and compare the amount and frequency of punishment between a delegated and non-delegated decision conditional on the outcome.

For RQ 2, we will look at the delegation frequency between the treatments (punishment possible vs punishment not possible).

4) How many and which conditions will participants be assigned to?

There are two conditions (punishment possible/no punishment possible) and two roles in each condition (delegator/evaluator).

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

I will run parametric (t-test) and non-parametric (MWU, Kolmogorov-Smirnov) tests between punishment (amount and frequency) for delegated and non-delegated outcomes, conditional on the outcome. Similarly, I will run linear regressions with punishment behavior as the dependent variable and explore whether expected performance or any socio-demographic characteristics are predictive of differential punishment.

For delegators, I will again, run parametric (t-test) tests on delegation choices between treatments. Moreover, I will employ Probit regressions with the dependent variable being the delegation decision and a set of explanatory variables including treatment, beliefs about performance and socio-demographic characteristics.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

I will exclude people that access the experiment via mobile device. Moreover, I may exclude participants that failed the attention checks.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

I will recruit subjects via Prolific. I plan to recruit 320 participants, 160 per treatment, 80 per role per treatment.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

I already ran a pilot with 25 observations in the punishment treatment. I plan to include the data in the analysis.

Secondary analysis will include an investigation on how performance and beliefs about performance impact results. Moreover, I will look at the delegator's beliefs about expected punishment and their consistency with the delegation decision.

At this point, I already want to state that I may run another treatment in which the delegator may either delegate to another participant or an algorithm.